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Paleomagnetic and $^{40}Ar/^{39}Ar$ geochronological results from the Linzizong Group, Linzhou Basin, Lhasa Terrane, Tibet: Implications to Paleogene paleolatitude and onset of the India–Asia collision

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ABSTRACT

The Linzizong Group (64–44 Ma) of the Lhasa Terrane in Tibet is critically positioned for establishing the paleoposition of the southern leading edge of the Asian continent during Paleogene times and constraining onset of the India–Asia collision. Here we report paleomagnetic results from a collection comprising 384 drill-core samples from 34 sites embracing all three formations of this group. Comprehensive demagnetization and field tests isolate characteristic remanent magnetizations (ChRM) summarized by overall tilt-corrected formation-mean directions of $D = 183.6°$, $I = -12.4°$ ($\alpha_{95} = 8.1°$) for the Dianzhong (64–60 Ma), $D = 1.0°$, $I = 18.1°$ ($\alpha_{95} = 8.1°$) for the Nianbo (60–50 Ma), and $D = 12.4°$, $I = 23.2°$ ($\alpha_{95} = 7.3°$) for the Pana (50–44 Ma). Fold tests are positive in each formation suggesting a pre-folding origin and we interpret the magnetizations as quasi-primary and acquired at, or slightly later than, formation of the Linzizong Group. Revised Paleogene paleopoles with Ar–Ar age constraints for the Lhasa Terrane indicate that onset of the India–Asia collision occurred no later than ~60.5 ± 1.5 Ma at a low paleolatitude of ~10°N. Analysis of 60 site-mean observations from a range of studies of the Pana Formation in the higher part of the succession highlight a large dispersion of ChRM directions; a number of possible causes are suggested but further study of this formation over a wider area is required to resolve this issue.

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1. Introduction

The collision between the Indian Plate and the collage of Asian terranes occurred in an approximately north–south direction. Since this orientation is most sensitive to paleomagnetic analysis a direct comparison of paleolatitudes from either side of the suture can potentially provide crucial evidence for the timing of collision. The Linzizong Group of Paleogene age developed in the Lhasa Terrane is spatially and temporally placed to provide a key constraint on this timing. Several recent paleomagnetic studies on this syn- or post-collisional volcanic and interbedded sedimentary sequence (Mo et al., 2003, 2006; Dong et al., 2005) have yielded a paleolatitude of less than ~22°N for the southern leading edge of the Asian continent during formation of this Group (Chen et al., 2010; Dupont-Nivet et al., 2010; Liebke et al., 2010; Sun et al., 2010; Huang et al., 2013) and appear to rule out an India–Asia collision in the Oligocene (c.f. Aitchison et al., 2000, 2007; Aitchison and Ali, 2001; Ali and Aitchison, 2008). Meanwhile, a paleomagnetic study of ~38–33 Ma volcanic rocks from southern Qiangtang in central Tibet predicted that the ~35 Ma paleolatitude for the southern margin of the Lhasa Terrane was 24.7 ± 3.7°N, given that the post-~35 Ma N–S directed upper crustal shortening within southern Tibet at the longitude of Lhasa is less than 100 km (Lippert et al., 2011). More recently, a comprehensive review on the 110–50 Ma paleomagnetic data of the Lhasa Terrane proposed a quasi-stationary paleolatitude of ~20°N (van Hinsbergen et al., 2012; Lippert et al., 2014). This estimate is, however, ~10–15°N higher than the one resolved from the lower part of the Linzizong Group (Najman et al., 2010), in which paleolatitudes for the southern margin of the Lhasa Terrane yielded ~6.7°N for the Dianzhong Formation (Chen et al., 2010), ~11.6°N for the Nianbo Formation (Chen et al., 2010; Sun et al., 2010), and ~13.2°N for dykes intruded into the Nianbo Formation (Liebke et al., 2010, 2012). This discrepancy in paleolatitudes might result from (i) complexities of paleomagnetic data from the Pana Formation, (ii) inadequate sampling of the Dianzhong and Nianbo formations from the Linzhou Basin,

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(iii) potential northward movement of the Lhasa Terrane during formation of the Linzizong Group, and (iv) potential recording biases such as inclination shallowing and/or remagnetization. To test these possibilities further study of the lower part of the Linzizong Group is required.

Our reported paleomagnetic reconnaissance of the Linzizong Group (Chen et al., 2010) produced a combined paleopole for the Nianbo and Pana formations (~60–44 Ma) but limited sampling notably from the middle and upper parts of the group precluded a tight constraint on motion of the southern leading edge of the Asian continent during early stages of the India–Asia collision. The key uncertainty surrounding the paleolatitude of the Lhasa Terrane during formation of the upper part of the Linzizong Group, and hence estimation of post-collisional crustal shortening within the Asian continent, remained. To help resolve this issue we report individual paleomagnetic poles for each of the three formations comprising the Linzizong Group in the Linzhou Basin of the Lhasa Terrane. Study of 384 new drill-core samples from the group enables us to constrain motion of the Lhasa Terrane during the interval ~64–44 Ma and permit further clarification of the timing of the India–Asia collision.

2. Paleomagnetic sampling

For the field sampling we follow 1:50,000 regional geological mapping of volcanic rocks in the Linzhou Basin by Dong et al. (2005) and focus on the area where the Linzizong Group was originally defined and further subdivided into three members comprising in ascending order the Dianzhong (~64–60 Ma), Nianbo (~60–50 Ma), and Pana (~50–44 Ma) formations. The lowest member of the Linzizong Group was sampled at 12 sites (123 drilled cores) in the vicinity of Kala village (3 sites, xl145–147) and Chongga village (9 sites, xl148–156) (Fig. 1 and Table 1). The lithologies of the sampled units in the middle part of the formation comprise mainly andesites which include distinguishable marker levels to resolve cooling cycles and bedding attitudes. The Nianbo Formation was sampled at 4 sites (xl157–160, Table 2) in the vicinity of the Namugunba Temple where they comprise ignimbritic units. The Pana Formation was sampled at 18 sites (xl127–144) in the vicinities of Nianbo village and Taju hill (Fig. 1); the sampled rocks comprise tuffs, ignimbrites, and tuffaceous andesites (Fig. 1 and Table 3). In total, 384 oriented drill samples from 34 sites have been collected from the Linzizong Group in the Linzhou Basin with the aim of incorporating the results with the earlier data from the same basin as reported by Chen et al. (2010). In addition three block samples were collected for $^{40}Ar/^{39}Ar$ dating, one located at the base of the Dianzhong Formation (xl156) and the other two from the base (xl127) and top (xl144) of the sampled section in the Pana Formation. We note that He et al. (2007) propose a different subdivision of the Linzizong Group in the sampling basin consisting of four subunits and this subdivision is also shown for comparison in the simplified geological map by the dotted lines (Fig. 1).

All cores were drilled with a portable petrol-powered drill and oriented mostly by sun compass. Some cores were oriented by both magnetic and sun compasses and the average difference between sun and magnetic compasses readings was $-0.20 \pm 0.64°$ ($n = 29$, 2σ). This difference is generally consistent with local declination calculated from the International Geomagnetic Reference Field (IGRF) model (~359.8°) for the sampling locality at 29.99°N, 91.14°E. Thus magnetic compass readings could be used without any corrections in this study.

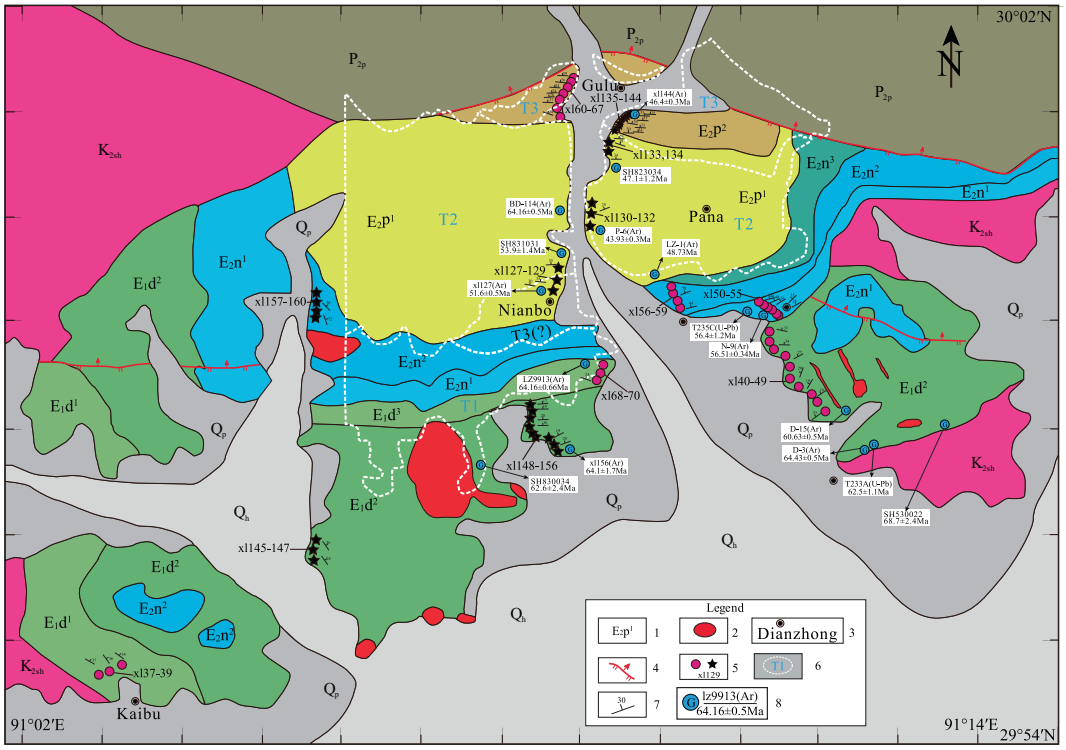


Fig. 1. Simplified regional geological map of the Linzhou Basin showing the distribution of paleomagnetic and geochronological sampling sites. For comparison with the stratigraphic subdivision of the Linzizong Group in the basin, the stratigraphic subdivision by He et al. (2007) is also shown in the plot. *Legend:* 1, stratigraphic units: P_{2p}, Upper Permian Pangna Formation; K_{2sh}, Low Cretaceous Shexing Formation; E_1d^1, E_1d^2 and E_1d^3 indicate lower, middle and upper parts of the Paleogene Dianzhong Formation; E_2n^1, E_2n^2 and E_2n^3 indicate lower, middle and upper parts of Eocene Nianbo Formation; E_2p^1 and E_2p^2 represent lower and upper parts of Eocene Pana Formation; Q_h and Q_p represent the Holocene and Pleistocene; 2, intrusive unit; 3, villages 4, thrust fault; 5, paleomagnetic sites of Chen et al., 2010 (circles) and this study (stars); 6, stratigraphic division proposed by He et al. (2007); 7, bedding attitudes; 8, geochronological sampling layers and dating results of Zhou et al. (2004) and this study.

Table 1
Summary of paleomagnetic results of the ChRM from the Dianzhong Formation, Linzhou Basin of the Lhasa Terrane.

Site ID	Lithology	λ_s (°N)	φ_s (°E)	Strike/dip	n/n_0	R/N	D_g (°)	I_g (°)	D_s (°)	I_s (°)	κ	α_{95} (°)	λ_p (°N)	φ_p (°E)
Kala village profile, Linzhou Basin, Lhasa														
xl145	Andesites	29.9457	91.0946	326/24	7/8	7/0	164.0	−40.9	180.0	−30.2	298.3	3.5	−76.3	91.1
xl146	Andesites	29.9457	91.0946	326/24	7(2)/7	7/0	168.0	−36.8	181.0	−25.0	180.8	4.5	−73.2	87.7
xl147	Andesites	29.9458	91.0948	327/26	8/8	8/0	160.0	−29.4	172.0	−20.6	237.6	3.6	−69.3	113.9
Chongga village profile, Linzhou Basin, Lhasa														
xl148	Andesites	29.9565	91.1405	248/14	7(2)/7	3/4	333.7	6.2	333.7	−7.8	31.6	10.9	47.8	312.3
xl149	Andesites	29.9566	91.1405	248/14	5/7	0/5	13.6	2.2	14.1	−9.2	37.2	12.7	52.9	247.4
xl150	Andesites	29.9559	91.1356	268/16	8(2)/8	8/0	198.0	−21.6	196.0	−6.5	47.7	8.1	−59.3	58.5
xl151	Andesites	29.9557	91.1406	255/19	5/9	5/0	204.0	−10.6	203.3	4.3	187.6	5.6	−50.9	52.3
xl152	Andesites	29.9554	91.1405	255/19	9/9	9/0	205.0	−9.9	204.5	4.8	69.9	6.2	−50.1	50.9
xl153	Andesites	29.9554	91.1404	260/20	6(4)/7	0/6	1.7	7.2	1.8	−12.3	48.7	9.7	53.8	268.1
xl154	Andesites	29.9554	91.1398	273/16	8(2)/9	8/0	172.0	−43.2	174.0	−27.4	282.6	3.3	−73.6	112.2
xl155	Andesites	29.9555	91.1394	273/16	7/8	7/0	162.0	−19.7	164.0	−4.7	113.1	5.7	−58.5	123.0
xl156	Andesites	29.9550	91.1392	273/16	7/7	7/0	180.0	−15.5	180.1	0.5	500.5	2.7	−59.8	90.9
Sub-mean					12/12	9/3	179.2	−21.1			13.6	12.2		
									182.2	−6.4	14.1	12.0		
Nagagunba-Xiagunba profile and Chongga profile, Linzhou Basin, Lhasa (Chen et al., 2010)														
xl40	Andesites	29.9588	91.1960	234/33	5(1)/6	5/0	190.3	−45.0	176.7	−19.1	35.1	13.1	−69.6	100.6
xl41 + 42	Andesites	29.9588	91.1960	236/19	7(5)/14	6/1	181.7	−51.0	172.6	−34.6	27.1	11.8	−77.2	124.5
xl43[a]	Andesites	29.9588	91.1960	236/19	5/6	1/4	14.0	51.0	1.6	36.7	16.0	19.7	−80.4	82.2
xl44[a]	Andesites	29.9598	91.1960	210/19	2/5	0/2	240.6	4.8	240.6	−5.0	–	–	26.5	194.6
xl45	Tuffaceous andesites	29.9601	91.1909	190/29	5(1)/9	5/0	184.5	−22.8	173.8	−17.3	30.0	14.2	−68.1	107.8
xl46	Andesites	29.9645	91.1960	268/34	7/7	7/0	191.6	−52.9	186.6	−19.5	121.4	5.5	−69.1	72.7
xl47[a]	Andesites	29.9648	91.1956	268/34	3/7	3/0	176.1	37.0	173.3	71.0	93.8	12.8	−4.4	95.0
xl48	Andesites	29.9645	91.1836	268/34	6/6	6/0	187.8	−42.7	185.3	−9.1	144.1	5.6	−64.1	79.0
xl49	Andesites	29.9645	91.1836	282/31	8/8	8/0	195.4	−46.5	194.4	−15.6	92.1	5.8	−64.2	56.8
xl68 + 69	Tuffs	29.9617	91.1400	211/21	5/7	5/0	204.0	−30.0	194.0	−25.6	60.7	9.9	−69.0	50.0
xl70	Tuffs	29.9617	91.1400	211/21[b]	7/8	7/0	211.0	−27.6	200.3	−25.6	98.9	6.1	−65.0	38.1
Sub-mean					8/11	8/0	193.9	−40.2			34.4	9.6		
									185.5	−21.1	43.8	8.5		
Group-mean					20/23	17/3	184.5	−29.1			13.8	9.1	66.4	262.5
									183.6	−12.4	17.3	8.1	K = 28.0, A_{95} = 6.3	

Fold test:
(1) Watson and Enkin's test (1993): the optimum concentration is achieved at 97.5 ± 15.8% unfolding.
(2) McFadden's test (1990): N = 20, *in-situ* ξ_2 = 8.947, tilt-corrected ξ_2 = 1.650, statistical threshold ξ = 5.207 at 95% confidence level, indicative of a positive fold test.
Abbreviations: site ID, site identification; strike/dip, strike azimuth and dip of bed; n/n_0, number of samples joined the calculation/demagnetized, number showing in the parentheses indicate number of remagnetization circles used; R/N, samples show reversed and normal polarity; D_g, I_g (D_s, I_s), declination and inclination of direction *in-situ* (after tilt-correction); λ_s, φ_s, latitude and longitude of the sampling site; λ_p, φ_p, latitude and longitude of corresponding virtual geomagnetic pole (VGP) in stratigraphic coordinates.
[a] Sites were discarded duo to significant deviations (>30°) with the mean directions or large uncertainties (α_{95} > 15°).
[b] The strike/dip for this site in Chen et al. (2010) was incorrect.

3. $^{40}Ar/^{39}Ar$ analytical methods and results

The samples for dating were crushed and sieved between 250 and 400 μm fractions and thoroughly washed with distilled water and acetone. Plagioclase, alkali feldspar and groundmass were carefully hand-picked using a binocular microscope to reduce any visible impurities following paramagnetic separation, and the feldspar was cleaned in 5% HF for 10 min to reduce melt inclusion and adhering glass. The groundmass was cleaned in 5% HNO_3 for 5 min to reduce alteration effects. Samples were wrapped in aluminum foil and irradiated together with Bern4M muscovite standards monitored in position B4 of the 49-2 reactor in Beijing for 24 h. The reference age for the Bern4M is 18.69 ± 0.36 Ma (Baksi et al., 1996; McDougal and Harrison, 1999).

Total fusion of standards and high-resolution incremental heating analyses of samples were performed in the $^{40}Ar/^{39}Ar$ Geochronology Laboratory of the Institute of Geology and Geophysics, Chinese Academy of Sciences, on a MM5400 mass spectrometer operating in a static mode. To reveal and mitigate effects of alteration and/or partial Argon loss, more than 17 incremental steps of heating were the preferred mode of analysis due to the internal reliability criteria provided by the age spectrum technique (McDougal and Harrison, 1999). Samples were degassed at 650 °C for 30 min before being incrementally heated in a double vacuum furnace to reduce air contamination and the gasses released during each step purified by 2 SAES NP10 getters (operated at 350 °C and 100 °C, respectively) before introduction into the mass spectrometer for Ar isotope determination. The ^{40}Ar, ^{39}Ar, ^{38}Ar, ^{37}Ar and ^{36}Ar isotopic abundances were determined through linear extrapolation at time zero of peak intensities and data were corrected for system blanks, mass discriminations, interfering Ca, K derived argon isotopes, and the decay of ^{37}Ar since the time of the irradiation. The decay constant used throughout the calculations is $(5.543 \pm 0.010) \times 10^{-10}$ a^{-1} (Steiger and Jäger, 1977). Detail of the analysis and data processing procedures are outlined in Wang et al. (2006) and Yang et al. (2008). Results of the $^{40}Ar/^{39}Ar$ experiments are plotted as age spectra and isotope correlation diagrams in Fig. 2 and all data are available in Supplementary Table S1.

The plagioclase from andesitic lava xl156 produces an almost concordant age spectrum except for last four steps which yield older ages (Fig. 2a). Twelve consecutive steps accounting for 86.3% of the total ^{39}Ar released, define a plateau age of 64.1 ± 1.7 Ma (2σ, MSWD = 0.9). An inverse isochron age of 62.4 ± 4.8 Ma (2σ, MSWD = 0.9), calculated from the same steps, is in agreement with the plateau age. The $^{40}Ar/^{36}Ar$ intercept of 296.5 ± 2.6 (2σ) is very close to the air ratio, indicative of no

Table 2
Summary of paleomagnetic results of the ChRM from the Nianbo Formations from the Linzhou Basin, Lhasa Terrane.

Site ID	Lithology	λ_s (°N)	φ_s (°E)	Strike/dip	n/n_0	R/N	D_g (°)	I_g (°)	D_s (°)	I_s (°)	κ	α_{95} (°)	λ_p (°N)	φ_p (°E)
Namgunba profile, Linzhou Basin, Lhasa														
xl157	Ignimbrites	29.9583	91.0900	342/32	5/7	0/5	352.3	32.9	8.7	22.4	33.2	13.5	70.0	245.4
xl158	Ignimbrites	29.9583	91.0900	335/29	6(2)/8	0/6	5.2	26.5	13.3	9.9	232.7	4.4	62.1	241.8
xl159	Ignimbrites	29.9583	91.0900	335/29	8/8	0/8	2.0	25.6	10.2	10.3	491.5	2.5	63.4	247.9
xl160	Ignimbrites	29.9583	91.0900	335/29	8/8	0/8	359.9	27.2	9.2	12.6	89.1	5.9	64.9	249.1
Sub-mean					4/4	0/4	0.0	28.1			195.0	6.6		
									10.4	13.8	171.6	7.0		
Xiagunba, Renjigang and Gulu profiles, Linzhou Basin, Lhasa (Chen et al., 2010)														
xl50	Tuffaceous sandstones	29.9800	91.1803	241/24	5(1)/8	5/0	193.0	−58.0	178.0	−38.0	113.9	7.2	−81.2	103.4
xl51	Tuffaceous sandstones	29.9800	91.1803	241/24	8(7)/8	0/8	357.3	33.8	353.1	11.9	95.4	5.7	65.2	287.7
xl52	Conglomerate tuffs	29.9796	91.1795	226/35	7(7)/8	0/7	5.2	35.6	354.6	9.9	44.9	9.1	64.5	283.8
xl53	Tuffs	29.9796	91.1795	248/26	8(2)/8	2/6	353.6	34.3	351.0	9.1	54.1	7.6	63.2	291.4
xl54	Sandstones, limestones	29.9796	91.1795	247/40	6(3)/8	0/6	14.5	38.8	5.4	4.7	85.2	7.3	61.9	259.7
xl55	Sandstones, limestones	29.9789	91.1793	286/33	8(3)/8	0/8	14.5	47.4	15.0	14.4	30.4	10.2	63.3	236.3
xl56	Tuffs	29.9761	91.1676	256/44	8/8	8/0	232.0	−84.4	173.0	−43.5	72.2	6.1	−82.3	146.3
xl57	Argillaceous limestones	29.9761	91.1676	256/44	8/8	8/0	201.0	−74.9	176.0	−33.1	139.8	4.7	−77.5	109.1
xl58[a]	Limestones	29.9761	91.1676	256/44	8/8	6/2	290.9	−83.7	173.9	−49.3	11.3	17.2	−84.7	184.8
xl59	Argillaceous limestones	29.9761	91.1676	249/70	5/8	2/3	11.0	81.9	343.0	13.1	55.1	10.4	61.7	309.0
Sub-mean					9/10	3/6	6.8	54.9			13.8	14.4		
									356.5	20.1	23.8	10.8		
Group-mean					13/14	3/10	4.3	46.3			13.3	11.8	69.7	268.6
									1.0	18.1	27.0	8.1	K = 43.6, A_{95} = 6.3	

Fold test:
(1) Watson and Enkin's test (1993): the optimal concentration is achieved at N = 13, 95.1 ± 15.1% unfolding level.
(2) McFadden's test (1990): N = 13, *in-situ* ξ_2 = 6.546, tilt-corrected ξ_2 = 3.625 statistical threshold ξ = 4.200 at 95% confidence level, indicative of a positive fold test.
The abbreviations are the same as for Table 1.

apparent excess argon contamination (Fig. 2b). As radiogenic ^{40}Ar percentages are relatively low in the total ^{40}Ar released in each step, the air correction leads to large uncertainty of apparent age as well as the final plateau age for sample xl156.

The alkali feldspar from ignimbrite lava flow xl127 also yields a nearly concordant age spectrum with several disturbed ages confined to low and high temperature steps (Fig. 2c). Twelve consecutive steps accounting for 88.4% of the total ^{39}Ar released, define a plateau age of 51.6 ± 0.5 Ma (2σ, MSWD = 1.2). The inverse isochron (Fig. 2d) for the same steps yields atmospheric initial $^{40}Ar/^{36}Ar$ (298.0 ± 3.7) and an age of 50.9 ± 1.1 Ma (2σ, MSWD = 1.1) in close agreement with the plateau age.

The groundmass from lava xl144 yields an age spectrum rising slightly with low ages at the beginning of the spectrum due to alteration (Fig. 2e). Twelve consecutive steps, accounting for 72% of the total ^{39}Ar released, define a plateau age of 46.4 ± 0.3 Ma (2σ, MSWD = 1.1). The inverse isochron (Fig. 2f) for the same steps yields atmospheric initial $^{40}Ar/^{36}Ar$ (295.4 ± 7.7) and an age of 46.4 ± 0.8 Ma (2σ, MSWD = 1.2) in close agreement with the plateau age. The low temperature steps yield meaningless ages evidently due to alteration effects.

The inverse isochron ages are generally more objective because they require no assumptions about the initial $^{40}Ar/^{36}Ar$ ratios and are recommended when the $^{40}Ar/^{36}Ar$ ratio is obviously different from the air ratio. However, all the samples here have normal $^{40}Ar/^{36}Ar$ ratios as the air and the plateau ages all have smaller uncertainties than the inverse isochron ages. Hence we interpret the plateau ages of 64.1 ± 1.7 Ma, 51.6 ± 0.5 Ma and 46.4 ± 0.3 Ma as defining eruption times of these three lava flows. These are in good agreement with ages for the Dianzhong and Pana formations reported by previous studies (Zhou et al., 2004; He et al., 2007; Lee et al., 2009; Fig. 1) and crucially help to provide a temporal framework for paleomagnetic results from the Linzizong Group (Sections 6.2 and 6.3).

4. Rock magnetic results

The collected samples have similar lithologies to the rock units previously studied by Chen et al. (2010) and rock magnetic properties prove to be comparable. In general, thermomagnetic spectra for purple–red ignimbrites from the Nianbo Formation (sites xl157–160) show approximately-reversible k–T curves with unblocking temperatures of ~650–670 °C indicating that hematite is the main magnetic mineral carrier. In the pink andesites, rhyolitic andesites, and tuffs from volcanic units of the Dianzhong and Pana formations, the majority of samples are characterized by irreversible k–T curves and show significant depression of magnetic susceptibility during heating up to 700 °C (Fig. 3a–d). Significant drop of magnetic susceptibility around ~580–590 °C and subordinate decrease of magnetic susceptibility above 600 °C indicate that these volcanic rocks are ferromagnetically dominated by magnetite and hematite. To further identify temperature dependence of magnetic susceptibility during heating, a few samples were subject to multiple heating–cooling cycles with step-by-step increase in the termination temperatures. As shown in Fig. 3e, the irreversible k–T behavior of sample xl130-5 (rhyolitic ignimbrite, Pana Formation) results from a sharp increase around 200 °C and a significant drop around 300–400 °C; little or no change is observed between the heating and cooling recycles up to 530, 590 and 700 °C. The first sharp variation of magnetic susceptibility between 200 and 300 °C may result from exsolution of titanomagnetite to magnetite, while the latter changes could indicate inversion or oxidation of maghemite to hematite for the surface partially-oxidized volcanic rocks (Dunlop, 1995). However, for another typical sample xl140-10 (ignimbrite, Pana Formation), similar variation of magnetic susceptibility is observed during heating–cooling up to 260 °C with no sharp drop of magnetic susceptibility around ~400 °C (Fig. 3f). Significant irreversible k–T behavior seems to occur between 500 and 600 °C, resulting in loss

Table 3
Summary of paleomagnetic results of the ChRM from the Pana Formation from the Linzhou Basin, Lhasa Terrane.

Site ID	Lithology	λ_s (°N)	φ_s (°E)	Strike/dip	n/n_0	R/N	D_g (°)	I_g (°)	D_s (°)	I_s (°)	κ	α_{95} (°)	λ_p (°N)	φ_p (°E)
Nianbo village profile (western side of the valley), Linzhou Basin, Lhasa														
xl127	Rhyoignimbrites	29.9881	91.1414	301/30	8(8)/8	8/0	200.8	−61.6	205.7	−31.9	237.6	3.6	−63.3	23.8
xl128	Rhyoignimbrites	29.9885	91.1411	294.5/33.3	8/8	8/0	226.0	−60.0	216.5	−27.9	300.4	3.2	−53.2	17.3
xl129	Ignimbrites	29.9886	91.1410	262/35	8/8	8/0	216.0	−63.5	193.8	−33.3	580.5	2.3	−72.8	41.3
Nianbo-Gulu valley profile (Taji Hill, eastern side), Linzhou Basin, Lhasa														
xl130[a]	Rhyoignimbrites (quarry)	30.0029	91.1486	249/29	7/8	0/7	90.8	80.0	359.9	63.2	267.0	3.7	75.3	90.9
xl131[a]	Rhyoignimbrites (quarry)	30.0029	91.1486	249/29	8/8	0/8	86.9	76.5	7.5	62.3	850.1	1.9	75.1	112.8
xl132[a]	Rhyoignimbrites (quarry)	30.0029	91.1480	249/29	6/8	0/6	69.0	80.7	357.6	59.7	139.1	5.7	79.3	81.3
					2/8	1/1	42.5	32.5	31.3	16.8	–	–	53.7	210.9
xl133	Rhyolites	30.0088	91.1502	277/38	8/8	0/8	35.7	82.8	11.9	45.6	300.4	3.2	79.1	194.1
xl134	Ignimbrites	30.0100	91.1502	270/38	7/7	0/7	87.9	76.9	20.5	49.7	159.1	4.8	72.3	174.3
xl135[a]	Ignimbrites	30.0104	91.1500	284/29	6(2)/7	1/5	5.3	33.3	6.7	4.6	11.8	23.2	61.6	257.0
xl136	Ignimbrites	30.0106	91.1500	276/29	7(1)/8	0/7	5.7	42.4	5.8	13.4	73.2	7.1	66.2	256.8
xl137	Ignimbrites	30.0108	91.1500	279/34	7/8	0/7	8.5	41.2	8.7	7.2	24.0	12.6	62.3	252.2
xl138	Ignimbrites	30.0109	91.1502	278/33	6(1)/8	0/6	34.9	59.9	23.0	28.9	187.9	4.9	64.3	210.7
xl139	Ignimbrites	30.0110	91.1502	281/34	8/8	0/8	19.8	54.5	16.5	20.7	63.6	7.0	65.3	229.2
xl140	Ignimbrites	30.0111	91.1501	279/34	7(2)/7	0/7	24.3	52.8	19.1	20.7	40.5	9.6	63.8	224.5
xl141	Tuffites	30.0111	91.1504	282/35	8/8	0/8	15.4	50.0	14.2	15.0	110.2	5.3	64.0	237.5
xl142	Tuffites	30.0114	91.1500	283/33	6/7	0/6	349.0	52.7	357.6	21.4	134.4	5.8	70.9	278.4
xl143[a]	Rhyolites	30.0118	91.1510	281/34	5/5	2/3	97.0	12.3	90.8	7.9	4.9	38.4	1.3	177.3
xl144[a]	Rhyoandesites	30.0121	91.1510	278/35	6/7	6/0	282.8	11.5	288.5	6.7	7.4	26.5	17.7	354.7
Sub-mean					12/18	3/9	20.7	59.7			25.1	8.8		
									15.8	26.6	27.4	8.4		
Gulu-Nianbo profile (western side of the valley), Linzhou Basin, Lhasa (Chen et al., 2010)														
xl60	Tuffs, tuffaceous andesites	30.0193	91.1411	272/29	6/7	0/6	3.5	34.2	3.2	5.2	413.0	3.3	62.4	264.2
xl61	Tuffaceous andesites	30.0193	91.1411	276/26	8/8	0/8	6.1	30.3	6.1	4.3	580.5	2.3	61.5	258.3
xl62	Andesites, tuffs	30.0193	91.1411	274/24	8/8	0/8	4.3	34.2	4.3	10.2	696.1	2.1	64.8	261.0
xl63	Tuffs	30.0186	91.1411	266/26	7/7	0/7	357.9	33.0	357.6	7.0	189.1	4.4	63.4	276.5
xl64[c]	Conglomerate tuffs	30.0122	91.1403	327/22	8/8	0/8	7.1	51.6	21.4	35.2	40.6	8.8	67.9	204.9
xl65[a,c]	Conglomerate tuffs	30.0122	91.1403	327/22	8/8	0/8	293.1	79.4	27.9	71.8	45.5	8.3	57.0	119.3
xl66	Tuffs	30.0114	91.1403	272/37	6/9	0/6	12.2	72.4	5.8	35.6	100.9	6.7	78.4	242.8
xl67[c]	Tuffs	30.0101	91.1403	289/24	6/6	0/6	329.6	85.6	11.7	63.0	718.7	2.5	72.8	120.5
Sub-mean					6/8	0/6	4.2	42.5			23.8	14.0		
									5.9	16.3	23.9	14.0		
Group-mean					18/24	3/15	13.7	54.2			19.3	8.1	69.1	234.2
									12.4	23.2	23.5	**7.3**	$K = 38.7$	$A_{95} = 5.6$

Fold test: McFadden's test (1990): $N = 18$, *in-situ* $\xi_2 = 5.739$, tilt-corrected $\xi_2 = 1.059$ statistical threshold $\xi = 4.940$ at 95% confidence level, indicative of a positive fold test. The abbreviations are the same as for Table 1.
[c] Sites were discarded due to a significant collapse of the sampling strata in the original paper (Chen et al., 2010).

of magnetic susceptibility during the cooling; this is likely to be due to the partial oxidation of maghemite to hematite.

Anisotropy of magnetic susceptibility (AMS) was determined for 124 volcanic rock specimens. The corrected anisotropy degrees (P_j) are very low with a majority less than 1.04 and with a prevailing tendency for oblate shapes (Fig. 4a and b). The principal anisotropy directions are dispersed with a weak tendency for the k_{min} axes to be oriented perpendicular to the bedding plane both before and after tilt correction (Fig. 4c–n). The axial data sets are more closely grouped after tilt adjustment implying that these fabrics are essentially pre-tectonic. Application of tilt adjustment to the volcanic fabrics produces a girdle trending ~80°E with pole to plane dipping ~10°/175°E (Pana Formation, Fig. 4e and f) to ~10°/215°E (Nianbo Formation, Fig. 4i and j). In addition a deformation fabric would be expected to produce a contrasting E–W k_{max} direction perpendicular to the regional stress. Hence we prefer to interpret the fabric signature in terms of primary volcanic flow from a southerly-to-southwesterly direction with weak flow imbrication. The residual elongation in the grouped distributions may reflect incipient tectonic additions including possible unrecognized plunge effects (Fig. 4e–f, i–j, and m–n).

5. Paleomagnetic results

Following rock magnetic evaluation of pilot samples, ~8 specimens per site were subject to either progressive thermal or alternative field (AF) demagnetization. Demagnetization methods, techniques of analysis, and statistic treatment are reported in Chen et al. (2010 and references therein). In addition, a few paired volcanic specimens cut from same drill-core samples were subject to both progressive thermal and AF demagnetizations to determine whether component resolution was significantly influenced by magnetic mineral variation during the thermal treatment. All remanence measurements were performed on a 2G-755 cryogenic magnetometer in the Paleomagnetism and Geochronology Laboratory (PGL), Institute of Geology and Geophysics, Chinese Academy and Sciences (IGGCAS) where both demagnetizers and magnetom-

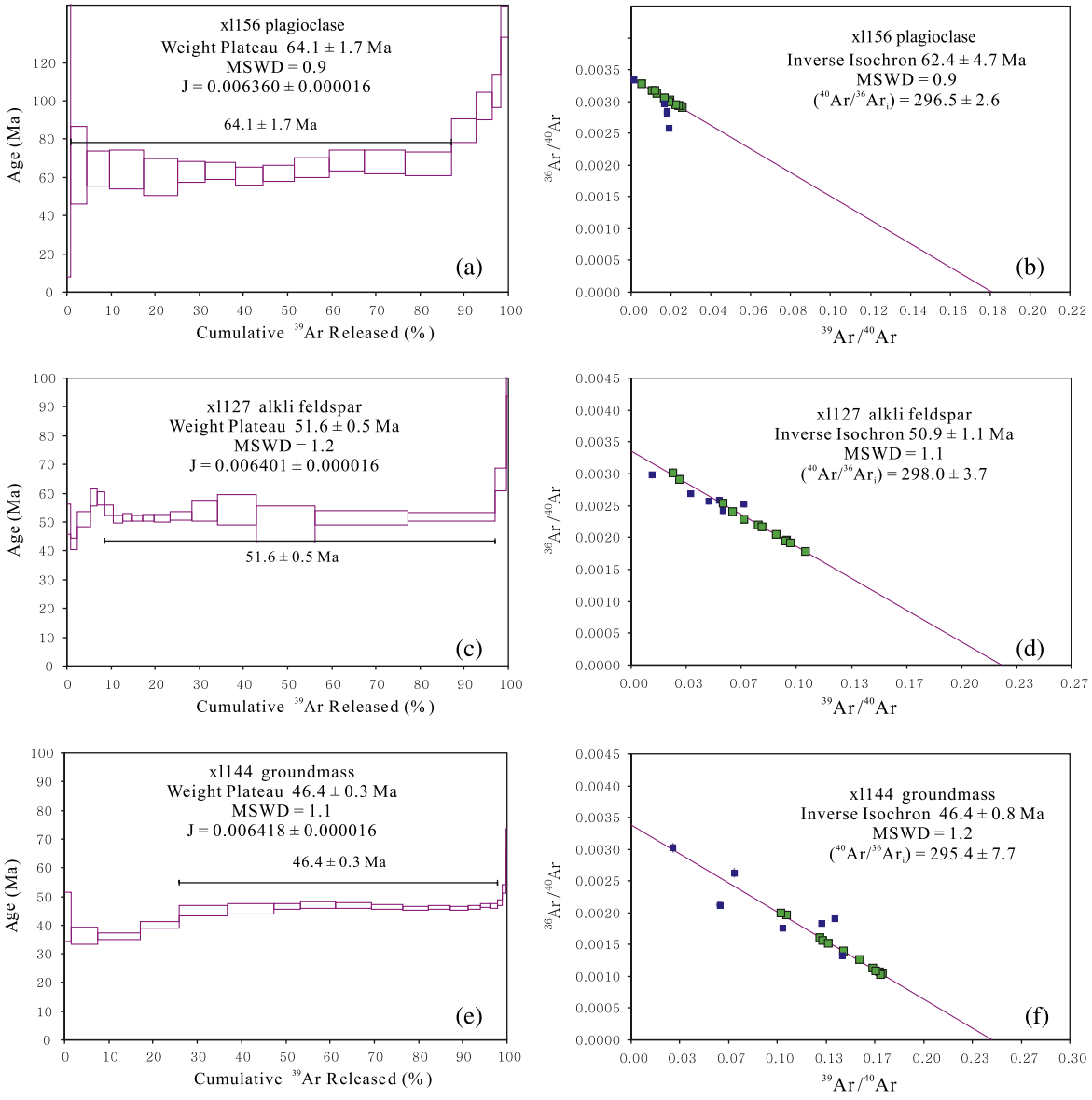


Fig. 2. Incremental heating age spectra and inverse isochron of the plagioclase (XL156), alkali feldspar (XL127), and groundmass (XL144).

eter are installed in a magnetically shielded space with the field inside minimized to less than 300 nT.

5.1. Dianzhong Formation

The majority of Characteristic Remanent Magnetizations (ChRMs) isolated by progressive thermal and AF demagnetization show a single component steadily decaying to the origin of orthogonal projections with unblocking temperatures of 550–590 °C (Fig. 5b and c). Specimens showing two magnetic components (Fig. 5d) comprise a low temperature (field) component, designated component A mostly removed by temperature steps of 250/300 °C or AF steps of 15/20 mT. The succeeding high temperature (field) component B is isolated by demagnetization temperatures (fields) above 350 °C (25–30 mT). In a few specimens high-temperature demagnetization trajectories mostly follow a great circle and permit isolation only by a remagnetization great circle between temperatures of ~350 °C and <670 °C (Fig. 5a), indicating that a ChRM overprinted by component A. The latter component has *in-situ* inclinations ~10° with slight deterioration of data grouping following tilt correction (*ks*/*kg* = 0.85) indicating that component A may have a post-folding origin and is therefore more likely to be a viscous overprint and/or recent remagnetization. The ChRMs (including component B) show directions distinct from component A; they are predominantly of reversed polarity with three sites (xl148, 149 and 153) showing normal polarity. The 12 site-mean directions yield a formation mean of $D = 179.2°$, $I = -21.1°$ ($\alpha_{95} = 12.2°$) before and $D = 182.2°$, $I = -6.4°$ ($\alpha_{95} = 12.0°$) after tilt correction. Combined with the results from the Linzhou Basin of Chen et al. (2010), a ChRM (B) is isolated from a total of 20 sites with direction and Fisherian mean of $D = 184.5°$, $I = -29.1°$ ($\alpha_{95} = 9.1°$) before and $D = 183.6°$, $I = -12.4°$ ($\alpha_{95} = 8.1°$) after tilt correction showing a slight improvement in directional grouping after tilt correction (*ks*/*kg* = 1.25, Fig. 6). The McFadden (1990) fold test is positive ($\xi_2 = 8.947$ *in-situ*, $\xi_2 = 1.650$ after tilt correction, and statistical threshold $\xi = 5.207$ at 95% confidence level) with optimal directional data grouping achieved at 97.5 ± 15.8% unfolding level and suggesting a pre-folding origin for component B.

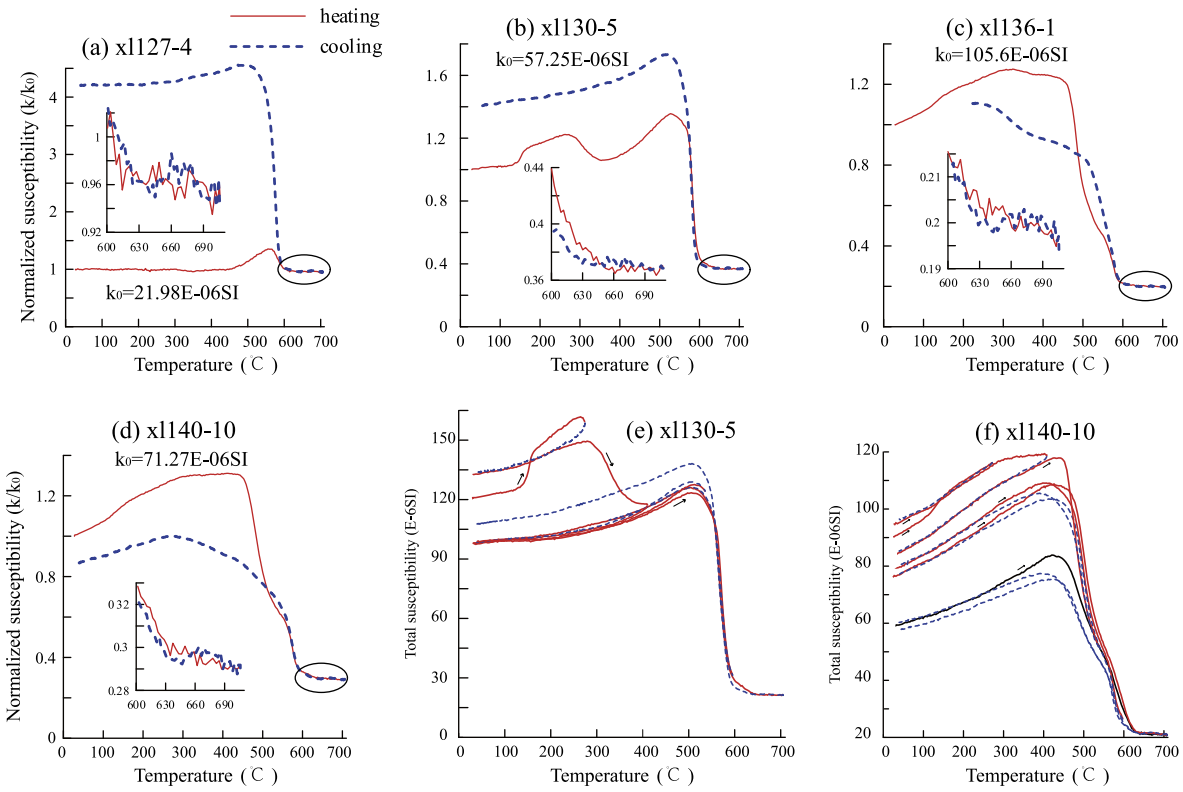


Fig. 3. Thermomagnetic behaviors of pilot volcanic rocks from the Dianzhong and Pana formations in the Linzhou Basin, Tibet. (a–d) Magnetic susceptibility versus temperature (k–T) curves in a single heating (solid lines) and cooling (dashed lines) run up to ~700 °C, and (e and f) k–T curves in multiple heating–cooling runs with stepwise increasing termination temperatures of 260, 400, 530, 590, and 700 °C. All measurements were performed in argon gas.

5.2. Nianbo Formation

Progressive thermal demagnetization of 31 ignimbrite specimens from 4 sites (xl157–160) successfully isolated a high-temperature ChRM in 27 specimens between temperatures of 300/350 °C and 575/590 °C (Fig. 5e) or between 400/450 °C and 670/690 °C (Fig. 5f) following removal of a viscous overprint in the initial stage of demagnetization. Two ignimbrite specimens from site xl158 exhibited a high-temperature demagnetization trajectory mostly following a great circle permitting component isolation by a remagnetization great circle between temperatures of ~400 °C and 670 °C; four of the remaining 6 specimens showed near-straightforward decay of the NRM intensity to the origin and yielded four component directions from principal component analysis (Kirschvink, 1980). Site mean direction of the ChRM at this site is calculated following the McFadden and McElhinny (1988) method (Table 2). The ChRMs isolated from the 4 newly-collected sites are all of normal polarity and data-grouping deteriorates slightly following tilt adjustment ($ks/kg = 0.88$) although this may have no significance. Combining with 9 site mean results of Chen et al. (2010) yields a total of 13 site mean ChRMs with a formation mean of $D = 4.3°$, $I = 46.3°$ ($\alpha_{95} = 11.8°$) before and $D = 1.0°$, $I = 18.1°$ ($\alpha_{95} = 8.1°$) after tilt correction (Table 2 and Fig. 7). The data grouping is therefore significantly improved by adjustment for folding with an optimal grouping achieved by 95.1 ± 15.1% unfolding. Application of the McFadden (1990) fold test yields a positive result at the 95% confidence level; the statistics ξ_2 is 6.546 before and 3.625 after tilt correction with a critical threshold of $\xi = 4.200$ at this confidence level implying that the isolated ChRM is of pre-folding origin.

5.3. Pana Formation

Both progressive thermal and AF demagnetization were applied to samples from this formation. Thermal demagnetization trajectories for specimens at sites xl128–129, xl134 and xl142–144 show a slight drop of remanence at temperatures below ~300/350 °C followed by a straightforward decay to the origin (Fig. 5i). These relatively simple demagnetization spectra permit clear identification of a high-temperature ChRM at temperatures between either ~550 °C and 655–690 °C in sites xl128–129 or ~300/350 °C and 590–620 °C in sites xl142–144. The AF demagnetization of specimens from site xl141 exhibits similar demagnetization behavior to thermal demagnetization for specimens from neighboring sites xl142–144, with a ChRM isolated during alternating fields 30–89 mT. However, thermal demagnetization for all specimens from site xl127 shows great circle migration on an equal-area projection (Fig. 5j) permitting a site-mean ChRM determination only from intersection of great circles (Table 3). Thermal demagnetization of samples from sites xl133, xl135–140 identifies both a low-temperature component (LTC) and a high-temperature ChRM: the LTC is mostly removed by temperatures of ~350–450 °C while the ChRM is generally isolated at temperatures between 400/450 °C and 550–590 °C (Fig. 5k), and between 450–510 °C and 640–690 °C in a few cases. Some specimens from sites xl135–136, 138, and 140 show great circle migration yielding site-mean directions calculated by a combination of principal component directional observation and great circles (Table 3). Thermal demagnetization of 24 rhyolitic ignimbrite specimens drilled from a quarry at sites xl130–132 (Fig. 8a) exhibits more complicated demagnetization trajectories. As shown in Fig. 5g and h, the high-temperature ChRM appears after removal of a LTC at temper-

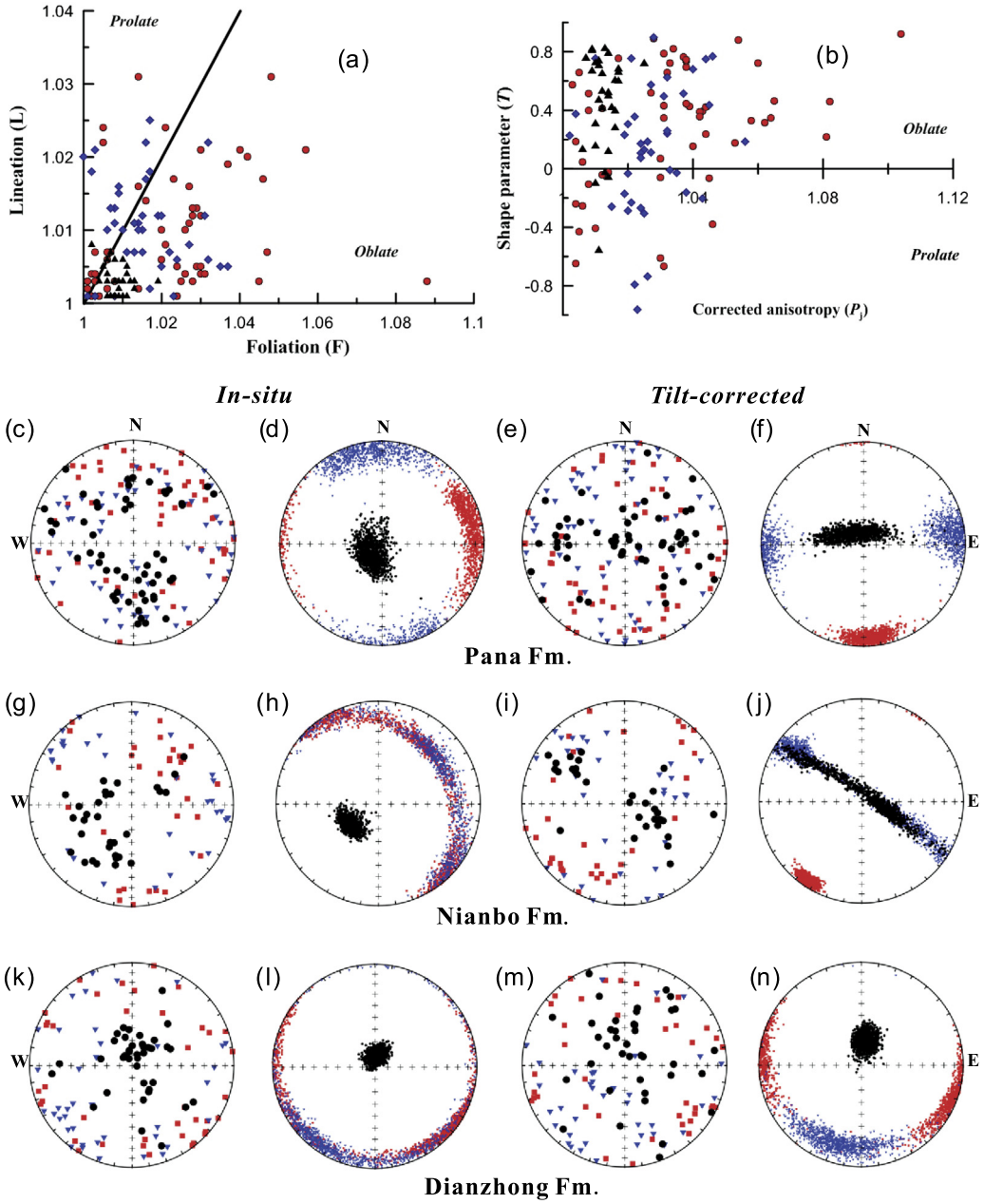


Fig. 4. Results from anisotropy of magnetic susceptibility measurements of 124 volcanic specimens collected from the Linzizong Group in the Linzhou Basin, Tibet. (a) Magnetic foliation versus lineation; (b) corrected anisotropy of magnetic susceptibility versus shape parameter; (c–n) lower-hemisphere projections of directions of maximum (squares), intermediate (triangles) and minimum (circles) principal axes of magnetic susceptibility ellipsoid before (c–d, g–h, and k–l) and after (e–f, i–j, and m–n) tilt correction; (d), (f), (h), (j), (l) and (n) equal-area projections of bootstrapped principal eigenvectors of 1000 pseudo-samples drawn from the data in (c), (e), (g), (i), (k) and (m).

atures below ~300 °C and an intermediate temperature component (ITC) by temperatures 480–510 °C. Consequently, the high-temperature ChRM was mostly isolated at temperatures between 530/540 °C and 640–670 °C.

The LTC could be isolated from 80 of 137 demagnetized specimens by at least 4 demagnetization endpoints up to temperatures of 300/350 °C. Following elimination of distinctly-dispersed directions, these LTCs yield a group-mean of $D = 4.4°$, $I = 48.3°$ ($\alpha_{95} = 3.3°$) *in-situ* and show slight deterioration of data grouping after tilt correction (Fig. 9a and b). We interpret this as a remagnetization in the Recent and/or Present Geomagnetic Field.

The ITC isolated by at least 5 demagnetization endpoints between temperatures of 300/350 °C and 480/510 °C could only be resolved in 24 specimens from sites xl130–132. These ITCs show a comparable directional distribution to high-temperature ChRMs isolated from the other sites except from xl130–132 either *in-situ* or after tilt correction. The HTCs isolated from 22 of these 24 are notably different from those isolated from the other sites in this formation (Figs. 8b and c, 9c and d & Table 3). These 22 HTCs have northerly moderate to high downward inclinations after tilt correction and exhibit uniform normal polarity. The remaining two specimens from site xl132 yield distinctive HTCs with north-east-northerly shallow downward inclinations of normal and reversed polarity after tilt correction (Table 3 and Fig. 8c), similar to high-temperature ChRM directions from all other sites in this formation; hence we suggest that the HTC may have been fixed by deuteric

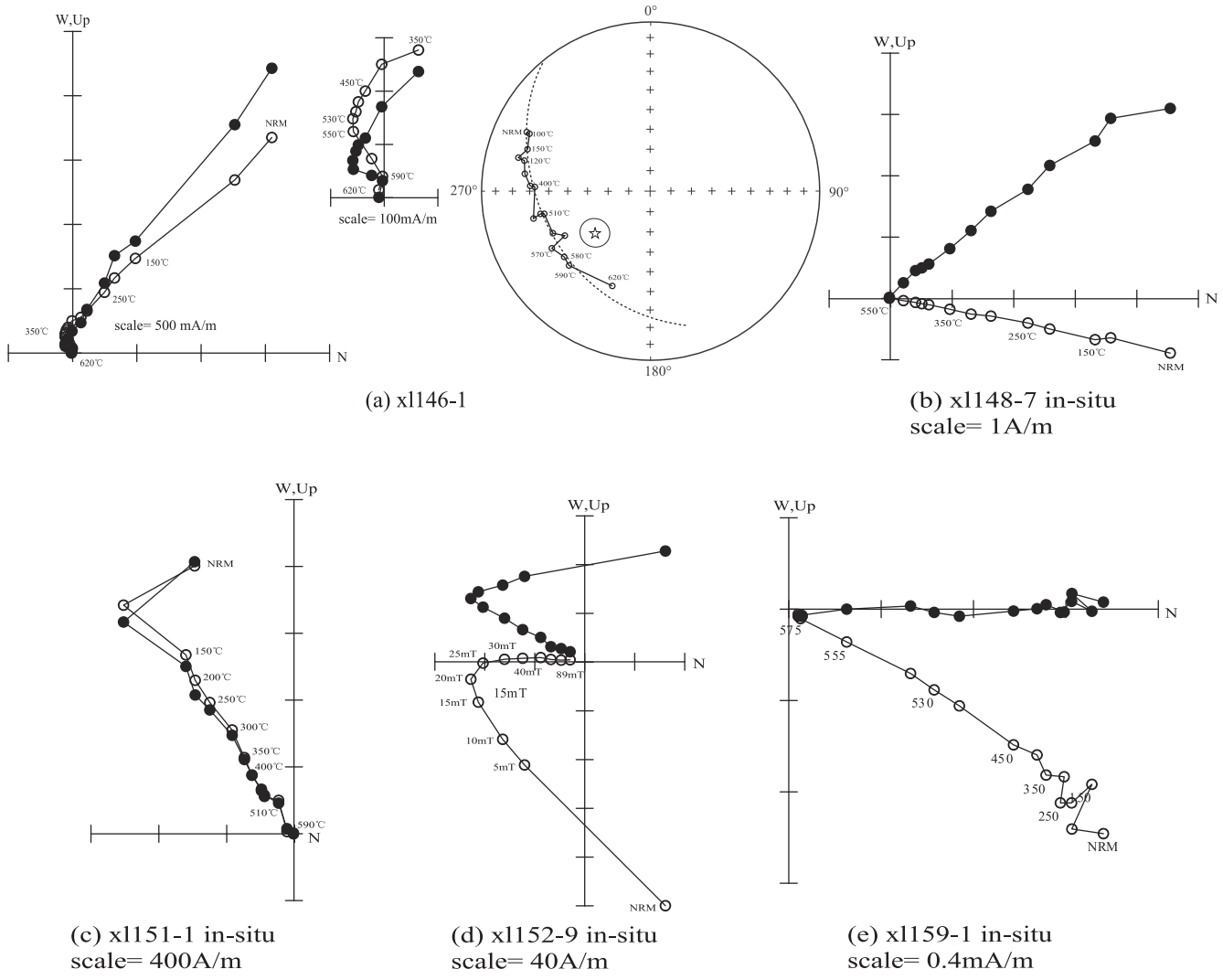


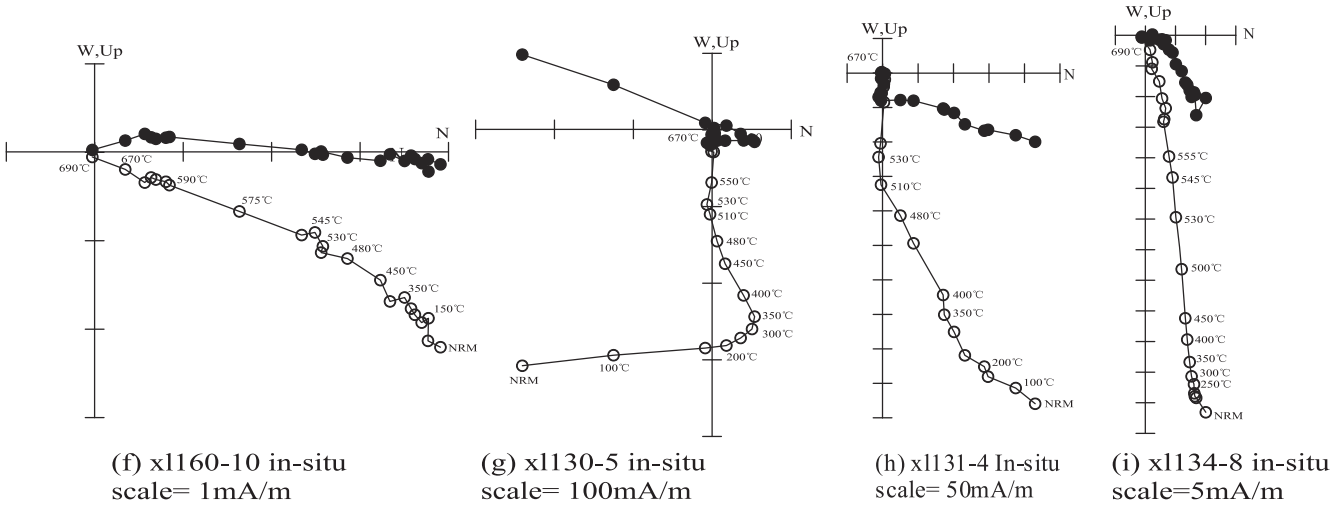
Fig. 5. Typical orthogonal and stereographic vector plots illustrating progressive thermal or AF demagnetization structures for Linzizong rocks from the Linzhou Basin. Orthogonal directions (Zijderveld, 1967) are plotted *in-situ*. Whereas stereographic directions are plotted in stratigraphic coordinates. Solid/open symbols represent endpoints projected onto horizontal or vertical planes, respectively.

oxidation of titanomaghemite shortly after lava eruption and records a polarity transition in an intermediate geomagnetic field direction. In view of this uncertainty we exclude the ITC and HTC from sites xl130–132 from further consideration.

Three sites (xl135, 143, and 144) yield site-mean directions with uncertainties greater than 20° and since these three sites all show mixed polarities suggesting that individual specimens might record the influence of transitional fields and we exclude them from further analysis. The high-temperature ChRMs from the remaining 12 sites comprise both normal and reversed polarity yielding a group-mean direction of $D = 20.7°$, $I = 59.7°$ ($\alpha_{95} = 8.8°$) before, and $D = 15.8°$, $I = 26.6°$ ($\alpha_{95} = 8.4°$) after, tilt correction with slight improvement in directional grouping ($ks/kg = 1.092$) following tilt adjustment. A reversal test (McFadden and McElhinny, 1990) yields a class C test ($\gamma_0 = 13.1°$) with a critical angle of $\gamma_c = 19.0°$ at 95% confidence level. Together with results reported by Chen et al. (2010), 18 site-mean observations produce a group-mean of $D = 13.7°$, $I = 54.2°$ ($\alpha_{95} = 8.1°$) before and $D = 12.4°$, $I = 23.2°$ ($\alpha_{95} = 7.3°$) after tilt correction (Fig. 9c and d) with maximum Fisherian precision achieved at complete 100% of unfolding. The McFadden (1990) fold test indicates a positive test result at 95% confidence level with the ξ_2 statistics of 5.739 before and 1.059 after tilt correction; the critical value of ξ is 4.940 at this confidence level.

6. Discussion

The paleomagnetic and geochronological results from the Linzizong Group in the Linzhou Basin reported in this study combined with results from previous investigation by Chen et al. (2010) permit the definition of three paleomagnetic poles from the Dianzhong (~64–60 Ma), Nianbo (~60–50 Ma), and Pana (~50–44 Ma) formations with revised positions of 66.4°N, 262.5°E ($N = 20$, $A_{95} = 6.3°$), 69.7°N, 268.6°E ($N = 13$, $A_{95} = 6.3°$), and 69.1°N, 234.2°E ($N = 18$, $A_{95} = 5.6°$) respectively (Tables 1–3). The robust quality of these formation-mean poles is reinforced by three tests: (i) Three formation-mean ChRM directions pass fold tests at the 95% confidence level with optimal directional concentrations around 100% complete unfolding (McFadden, 1990; Watson and Enkin, 1993). (ii) The ChRM from the Pana Formation possesses both normal and reversed polarity yielding a C class reversal test (McFadden and McElhinny, 1990). (iii) The three sample collections appear to have adequately sampled paleosecular variation (PSV) because angular standard deviations (ASDs, Vandamme, 1994) of the observed virtual geomagnetic poles (VGPs) are 15.4°, 12.3°, and 13.1° for the Dianzhong, Nianbo, and Pana formations respectively and consistent with distributions predicted from global PSV models during the last 5 Ma at latitudes of 10–30° (e.g. Johnson et al., 2008).



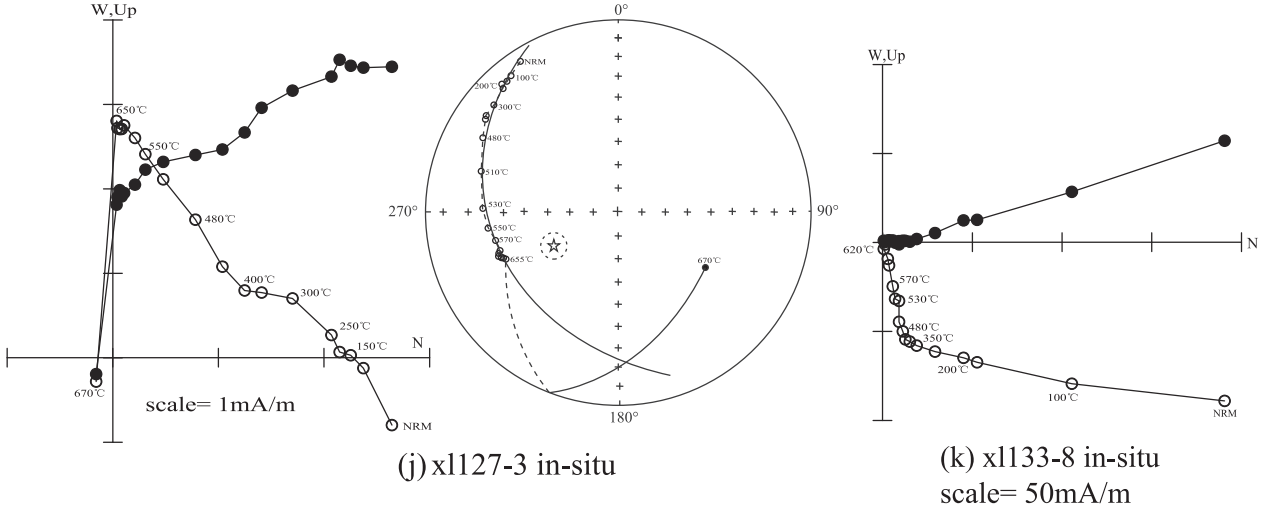


Fig. 5 (*continued*)

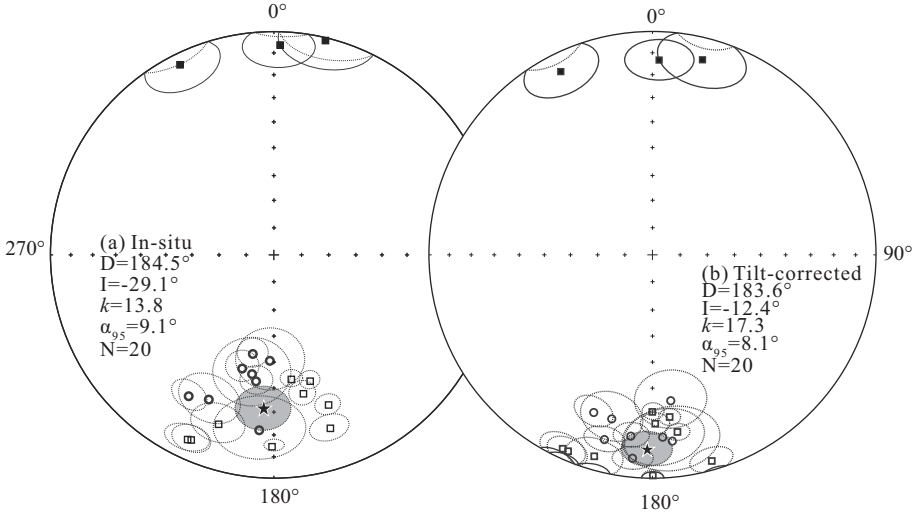


Fig. 6. Equal-area projections of directions of ChRM isolated from the Dianzhong Formation in the Linzhou Basin before (a) and after (b) tilt correction. Squares indicate site-mean directions of this study; circles indicate results of Chen et al. (2010); solid/open symbols represent positive/negative inclination; stars indicate Fisherian mean direction with 95% confidence limits.

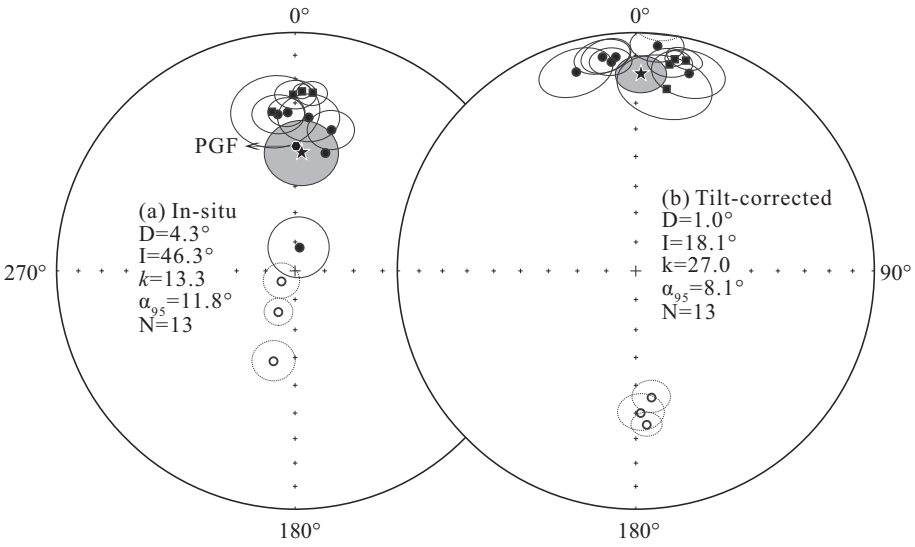


Fig. 7. Equal-area projections showing ChRM directions from the Nianbo Formation in the Linzhou Basin before (a) and after (b) tilt correction. Other symbols are as for Fig. 6.

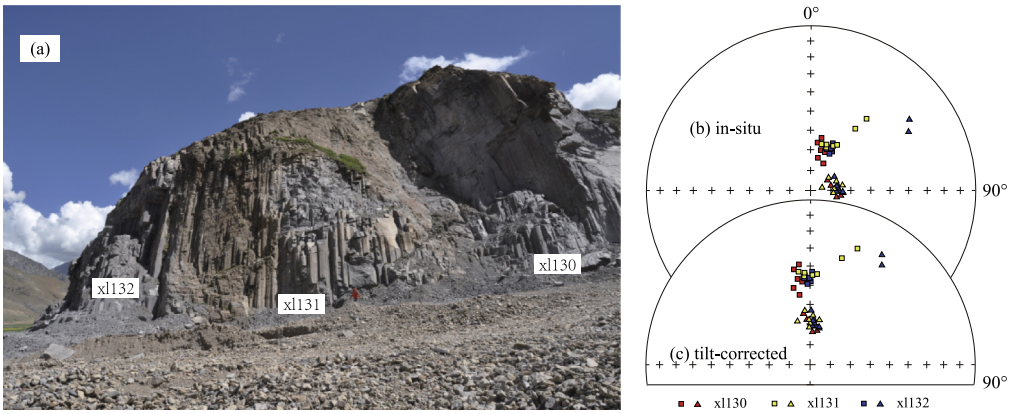


Fig. 8. (a) Photograph showing the middle part of the Pana Formation in the Linzhou Basin where sites xl130–132 were drilled from a quarry. (b and c) Equal-area projections of directions of the intermediate temperature components (ITC, squares) and high temperature components (HTC, triangles) isolated from specimens in sites xl130–132 before (b) and after (c) tilt correction. Other symbols are as for Fig. 6.

6.1. Paleoposition of the Lhasa Terrane during formation of the Linzizong Group

Since the bulk of collision-induced northward convergence between India and Asia occurred within a broad zone between mountain ranges of the Himalayan Front, the Tibetan Plateau and its northern hinterland, neither the relatively well-constrained Mesozoic–Cenozoic apparent polar wander (APW) path for Eurasia (e.g. Besse and Courtillot, 2002; Torsvik et al., 2008) nor Cretaceous and Paleogene paleopoles from Central Asia (e.g. Bazhenov and Mikolaichuk, 2002; Gilder et al., 2003; Huang et al., 2005) and Mongolia (Hankard et al., 2007, 2008; Metelkin et al., 2010) can be used to determine paleoposition of the southern leading edge of Eurasia (the Lhasa Terrane) during the critical Late Cretaceous–Paleogene interval. A paleogeographic analysis of the crust within this broad zone of deformation requires specific paleomagnetic information from the constituent terranes.

The newly-updated formation-mean paleomagnetic poles for the Linzizong Group define successive paleolatitudes of 6.6 ± 6.3°N (~64–60 Ma), 9.7 ± 6.3°N (~60–50 Ma) to 12.7 ± 5.6°N (~50–44 Ma) for the Lhasa Terrane at a reference site 30°N/91°E. Although these results indicate a non-significant northward motion of 6.1 ± 6.7° between formation of the lower and upper members of the Linzizong Group, the individual results are consistent with a successive northward slow motion at a rate of ~30 mm/yr of the Lhasa Terrane between ~64 and 44 Ma and suggest that the Lhasa Terrane was not precisely stationary during the Danian–Lutetian interval.

Several high-quality paleopoles have now been reported from the Lhasa Terrane specifically from the (Paleogene, ~64–44 Ma) Linzizong volcanic rocks and interbedded sediments, developed especially in the Linzhou Basin NE of the city of Lhasa (e.g. Dupont-Nivet et al., 2010; Liebke et al., 2010; Sun et al., 2010; Huang et al., 2013). On the basis of a wide analysis of the 110–50 Ma paleomagnetic data from the Lhasa Terrane, however, some researchers have proposed that the southern margin of Asia was stationary and located at ~20°N between ~110 and 50 Ma and including the interval of deposition of the Linzhou succession (e.g. van Hinsbergen et al., 2012; Lippert et al., 2014). This is at least ~10°N higher than our estimate from the lower part of the Linzizong Group.

This conflict highlights several problems with the ~110–50 Ma compilation. (i) Firstly, it discarded the data from the lower part of the group (the Dianzhong and Nianbo formations) suggesting the lower paleolatitude for the Lhasa Terrane on the grounds of inadequate sampling, remagnetization and inclusion of a sedimentary

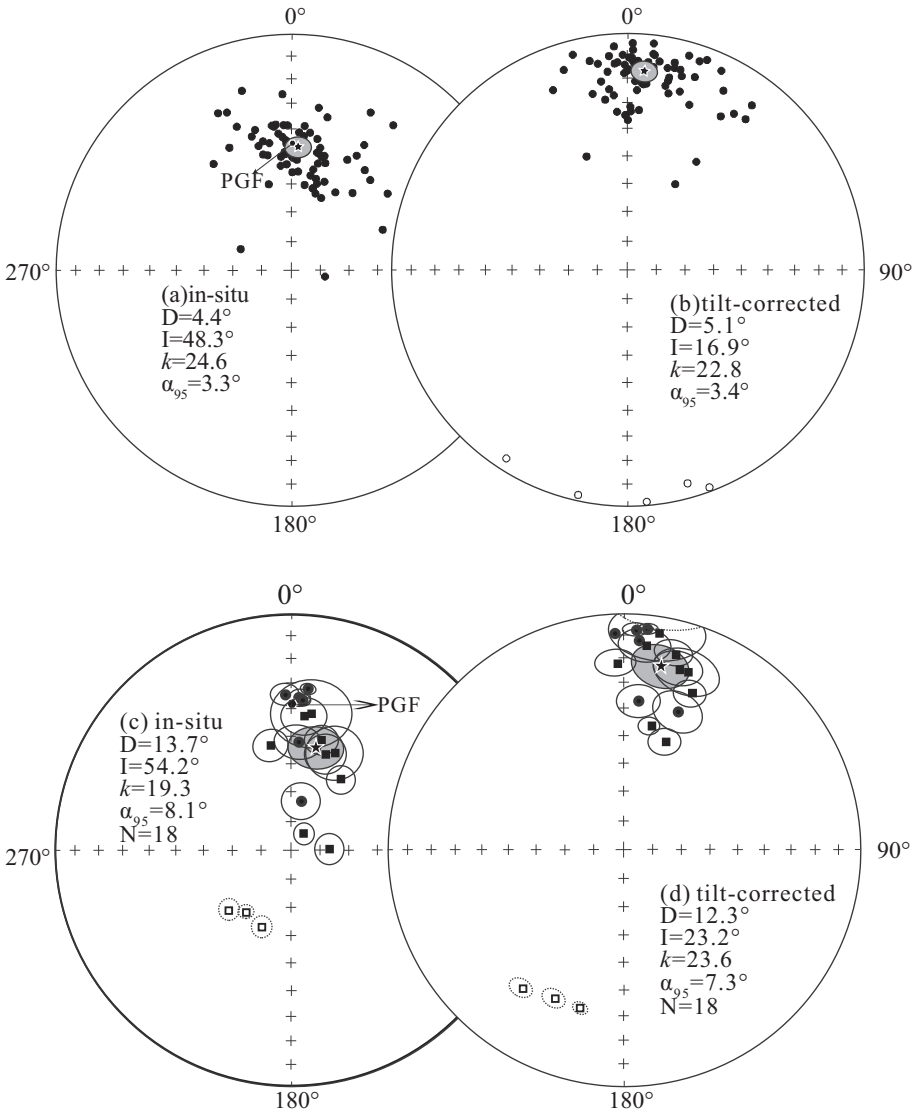


Fig. 9. Equal-area projections of site-mean directions of low temperature components (LTC, a and b) and high temperature characteristic remanent magnetizations (ChRM, c and d) isolated from the Pana Formation in the Linzhou Basin before (a and c) and after (b and d) tilt correction. Other symbols are as for Fig. 6.

dataset (van Hinsbergen et al., 2012; Lippert et al., 2014). However, our newly-updated paleomagnetic dataset for the Linzizong Group has demonstrated the robustness of the paleopoles from the lower part of the group by positive fold tests, adequate sampling of the PSV and similar inclinations recorded in both volcanic and sedimentary rocks. (ii) Secondly, paleolatitudinal estimates for the Late Cretaceous paleomagnetic data were mainly calculated from an inclination-corrected dataset from (Late Cretaceous) sediments of Sun et al. (2012) and Tan et al. (2010). We argue that an inclination-corrected dataset should not be exclusively used for the Cretaceous Lhasa Terrane, because it is not yet clear that significant compaction-induced inclination-shallowing has occurred within the Lhasa Terrane. Thus Sun et al. (2008, 2012) reported two Cretaceous paleopoles from separated localities in the eastern Lhasa Terrane, one from a volcanic succession (114.2 ± 1.1 Ma) cropping out near Deqing and yielding a paleolatitude of 13.6 ± 6.9°N for a reference point at 29.9°N and 90.7°E (Sun et al., 2008). The other is from (Late Cretaceous) Shexing red beds and three andesite sites (dated 72.4 ± 1.8 Ma by SHRIMP U–Pb method) in the vicinity of Maxiang yielding a paleolatitude of 17.5 ± 6.8°N for the same reference point. The positive fold test and similarity between sedimentary and volcanic results encouraged these authors to argue that the sediments had suffered non-significant compaction-induced inclination-shallowing although this assumption was questioned for results from the same sedimentary sequences in the Linzhou Basin (Tan et al., 2010). Sun et al. (2012) found that the majority of published Cretaceous paleomagnetic results from the Lhasa Terrane are distributed along a well-defined small circle centered on the reference point. Although this is the signature of distributed rotational deformation, it yields a meaningful Cretaceous paleolatitude of 15.5 ± 5.7°N for the Lhasa Terrane. More recently, paleomagnetic study of a red bed and lava flow succession in the Linzhou Basin (30.1°N/90.9°E) has also yielded compatible ChRM inclinations in sediments and volcanic rocks (Zhiming Sun, personal communication May 2014). In this study Sun and his colleagues collected 12 sites from Late Cretaceous Shexing red beds and 10 sites from interbedded lava flows (ca. 70–74 Ma by K–Ar dating in Peking University). Ten red bed sites and ten lava flow sites isolated group-mean ChRM directions of D/I = 2.4°/19.4° (α_{95} = 8.8°) and D/I = 0.6°/22.9° (α_{95} = 11.8°), respectively supporting the view that at least some of the Cretaceous sediments in the Lhasa Terrane have experienced no significant compaction-

induced inclination shallowing. The Paleomagnetic pole of this study, at 71.0°N, 266.4°E with A_{95} = 5.3°, indicates that the Lhasa Terrane was located at ~11°N during the interval ~74–64 Ma. In addition, two more paleomagnetic and geochronological investigations on volcanic rocks near Cuoqin (ca. 93–99 Ma, Tang et al., 2013) and Naqu (ca. 110–127 Ma by Zircon U–Pb LA-ICP-MS dating, Zhenyu Li et al., personal communication May 2014) yield comparable lower latitudes of ~15°N for the Lhasa Terrane during the Cretaceous.

Collectively these results show that the southern margin of the Asian continent comprising the Lhasa Terrane occupied low paleolatitudes around 15°N with little or no motion during Cretaceous times. Comparison between this mean Cretaceous paleolatitude and the updated results from the Dianzhong and Nianbo formations of the Linzizong Group indicates that southward motion of the Lhasa Terrane of no more than a few hundreds of kilometers occurred during the Late Cretaceous period. This could reflect marginal southward motion of the Eurasian plate during the Late Cretaceous as defined by southward transport of 6–10° during the interval between 100–90 Ma and 80–70 Ma from the Eurasian APW path (Besse and Courtillot, 2002).

6.2. Paleolatitudes of the Lhasa Terrane during formation of the Pana Formation

The paleomagnetic investigations of this study and Chen et al. (2010), Liebke et al. (2010, 2012) and Sun et al. (2010) are all consistent in confirming that the Lhasa Terrane was located around 10 ± 5°N during formation of the Dianzhong and Nianbo formations (~64–50 Ma). This estimate is found to be compatible with other Cretaceous observations (Section 6.1). However, paleolatitudes observed from the Pana Formation, the upper member of the Linzizong Group, range from less than 15°N (Chen et al., 2010 and this study) to higher than 30°N (Tan et al., 2010). Dupont-Nivet et al. (2010) and Huang et al. (2013) report paleolatitudes between these limits of ~22°N. All 60 site-mean ChRM directions now isolated from the Pana volcanics show a distribution ranging from ~10° to larger than 65° (Fig. 10a). At least four possibilities might explain this large inclination scatter: (1) there may be age differences in the sampled rock units so that the data could record, in part, fast latitudinal motion of the Lhasa Terrane between ~50 and 44 Ma; (2) incorrect tilt corrections could apply to some sites especially where there are uncertainties in determining bedding attitudes of the volcanic rocks; (3) some site-mean directions, particularly in the sedimentary units, may have experienced significant compaction-induced inclination-shallowing; and (4) large PSV or transitional geomagnetic field directions may be recorded in the collective analysis of Fig. 10a.

Noting that formation of the Pana Formation in the Linzhou Basin extended over ~6 Ma (Zhou et al., 2004 and this study) and given that the Lhasa Terrane could by then have had a comparable northward motion rate to the Indian plate (e.g. ~50–100 mm/yr, Acton, 1999) south of the IYSZ, northward transport of ~300–600 km is equivalent to an inclination difference of less than ~10° between the lowest and uppermost levels of the Pana Formation. Since we find no significant time-dependence of inclination in the site-mean observations (Table 3), the age-difference inclination scatter within the sampling sites should be slight or negligible. Secondly, although incorrect tilt-corrections from uncertainties in measuring bedding attitudes in the Pana volcanics could result in some errors in the site-mean results, we do not think this is a substantial reason for the inclination scatter because different observers have found consistent bedding attitudes at the same sampling sections (e.g. sampling sites xl130–xl132, GS2, GS22, GL01–GL06, and PN01–10, Supplementary Table S2). Thirdly, the site-mean ChRM directions isolated from the Pana Formation (this study, Chen et al., 2010; Dupont-Nivet et al., 2010; Tan et al., 2010) show a significant N–S elongation distribution (Fig. 10a) compatible with statistical model TK03.GAD for PSV of the geomagnetic field (Tauxe and Kent, 2004). This suggests that the observed ChRM directions record PSV of the geomagnetic field rather than a compaction-induced inclination abnormality. While the directions do not show an expected Fisher distribution (Fisher, 1953), the VGP distribution is approximately circularly-symmetric (Fig. 10b). Because reversals are recorded in some sampling sites (Supplementary Table S2) and normal and reversed polarities are sometimes found in different specimens from a single site (e.g. site xl132, Table 3), we consider that the large inclinational distribution scatter in the observed site-mean directions is likely to result from PSV of the geomagnetic field with the possible inclusion of transitional or excursional field directions (Fig. 10a).

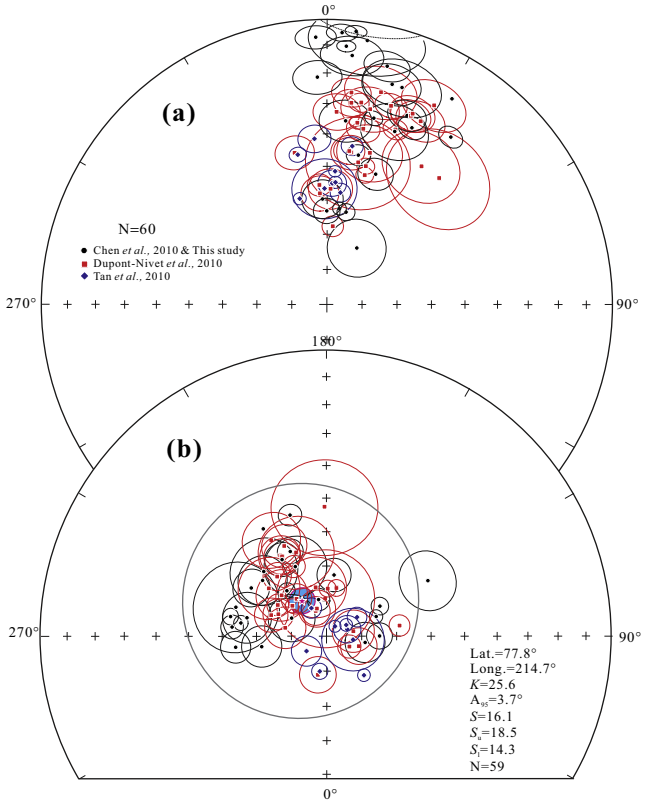


Fig. 10. Equal-area projections of tilt-corrected site-mean directions (a) and corresponding virtual geomagnetic pole (VGP) positions for all 60 sampling sites from the Pana Formation in the Linzhou Basin, Tibet. Square, diamond, and circle symbols indicate results from Dupont-Nivet et al. (2010), Tan et al. (2010), Chen et al. (2010) and this study, respectively. All symbols are plotted on the lower or northern hemisphere.

If this is the case the high implied paleolatitude (~32°N) for the Lhasa Terrane during formation of the Pana volcanics (Tan et al., 2010) may be an overestimate due to failure to average the PSV (Dupont-Nivet et al., 2010; Liebke et al., 2010; Lippert et al., 2014). An indication of this is the ASD of the VGPs of ~8.9° as determined by the Vandamme (1994) method which is significantly less than the expected ASD value at a paleolatitude of ~30° (Johnson et al., 2008). Consideration of all 60 site-mean observations from the Pana volcanics indicates that site xl65 (Chen et al., 2010) is likely to be transitional or excursional from a PSV cutoff angle of 34.0° (Vandamme, 1994). The remaining 59 VGPs yield a mean pole at 214.7°E, 77.8°N, K = 25.6, A_{95} = 3.7° with the ASD of ~16.1° (Fig. 10b) which is slightly larger than the expected ASD value at a paleolatitude of ~20° (Johnson et al., 2008) and indicates that other causes beyond the PSV of

geomagnetic field, such as the real northward transport of the Lhasa Terrane during formation of the Pana Formation as well as possible measurement errors in bedding attitudes of the Pana volcanics, could contribute to the inclination scatter of Fig. 10a. This mean result suggests a paleolatitude of about 23°N for the Lhasa Terrane during emplacement of the Pana volcanics and accords with the paleolatitude proposed by Dupont-Nivet et al. (2010). In addition, given that precision parameter *k*-value of 50 is considered to be a quality filter for paleomagnetic results isolated from volcanic rocks (e.g. Biggin et al., 2008; Deenen et al., 2011), 8 out of 60 site mean ChRM directions with *k*-values less than 50 should be discarded from the collection. However, the remaining 52 site mean ChRM directions do not yield significantly different Fisherian mean directions and corresponding paleolatitude estimates from the above analysis (Supplementary Table S2). The application of this filter does not therefore influence the comparison with other studies (e.g. Dupont-Nivet et al., 2010; van Hinsbergen et al., 2012).

We qualify this conclusion with the reservation that the above discussion is based upon equally weighting of the site-mean observations and it implies that all acceptable 59 site-mean observations are individually representative of the geomagnetic field during formation of the Pana Formation. It remains possible that some site observations temporally-overlap or that the geomagnetic field has been unevenly sampled, and consequently that this estimated paleolatitude is biased. Clearly the large inclinational dispersion of site-mean directions within the Pana volcanics is an outstanding problem, which will require further study based on sampling over a broader area.

6.3. Initial time of the India–Asia collision

The collective results from the Dianzhong and Nianbo formations of the Linzizong Group (~64–50 Ma) imply a low paleolatitude of 10 ± 5°N for the southernmost leading edge of the Asian continent (Chen et al., 2010; Liebke et al., 2010, 2012; Sun et al., 2010, and this study). Conservative assessment of deformation resulting from the India–Asia collision presupposes no paleomagnetically-detectable crustal shortening or thickening within the stable Indian Craton. Direct comparison between the updated Pana paleopole and the reference poles for the Indian plate at 50–45 Ma (Besse and Courtillot, 2002) indicates latitudinal flattening ($F \pm \Delta F$) of 4.87 ± 5.75° and 1.97 ± 6.28°, respectively and 8.85 ± 6.34° between the updated Nianbo paleopole and the reference paleofield direction for the Indian plate at 55 Ma. This implies a younger limit of ~50 Ma for the India–Asia collision.

Noting that pre-collisional Greater India has been constrained by paleomagnetic results from Paleocene marine sediments in Gamba and Duela, south Tibet (Patzelt et al., 1996; Yi et al., 2011), direct paleomagnetic comparison between Lhasa and Himalaya terranes indicates that onset of the India–Asia collision should have occurred prior to deposition of the magnetostratigraphically-studied Zongpu Formation (~62–56 Ma) in the Tethyan Himalaya. As shown in Fig. 11, paleolatitudinal flattening between the Late Cretaceous pole (~71–65 Ma, Patzelt et al., 1996) and the updated Dianzhong pole is relatively large at 10.43 ± 6.15° but the flattening reduces to become indistinguishable between the lower Zongpu pole (~62–59 Ma) and either the updated Dianzhong (−0.75 ± 5.77°) or Nianbo (2.35 ± 5.77°) poles. Using the upper Zongpu pole (~59–56 Ma) as a reference, we observe that paleolatitudinal transport between the Himalaya and Lhasa Terranes (using the updated Nianbo pole) is also paleomagnetically-indistinguishable (−2.00 ± 5.42°). Therefore the initial contact between India and Asia very likely occurred no later than ~60.5 ± 1.5 Ma at low paleolatitudes of ~10°N (Fig. 11). The consistency of this paleomagnetically-determined collision age with geophysical, geological and geochemical evidence for this collision is discussed elsewhere (Section 6.2 of Yi et al., 2011).

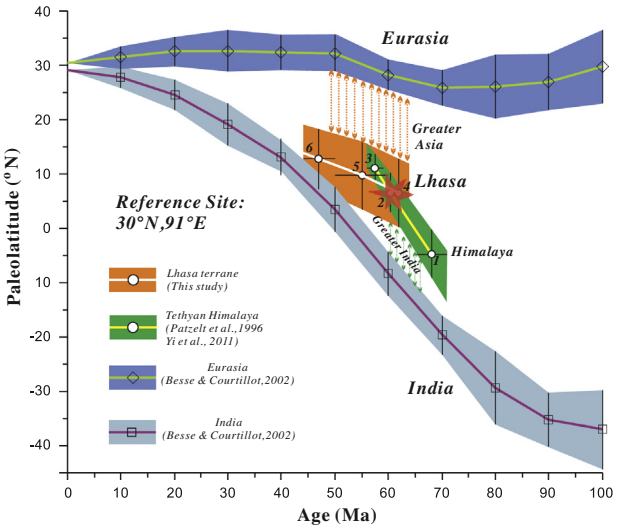


Fig. 11. Paleolatitudinal comparison of India, Eurasia, the Himalaya, and Lhasa during the post- Late Cretaceous times. The numbers 1–3 indicate average paleolatitudes for the Himalaya estimated from paleomagnetic results isolated from the Zongshan (Patzelt et al., 1996), lower Zongpu and upper Zongpu (Yi et al., 2011) formations; numbers 4–6 represent average paleolatitudes for the Lhasa Terrane during formation of the Dianzhong, Nianbo and Pana formations (this study). Paleolatitudes of Eurasia and India are computed from master apparent polar wander paths of Besse and Courtillot (2002).

7. Conclusions

An updated paleomagnetic study of the Linzizong Group in the Linzhou Basin of the Lhasa Terrane provides individual formation means for the three formations comprising this group and yields the following conclusions:

(1) The Ar–Ar age dates from the Dianzhong and Pana formations are compatible with previous studies (e.g. Zhou et al., 2004; He et al., 2007; Lee et al., 2009) confirming that the Linzizong Group was formed in the early Paleogene during the interval ~64–44 Ma. Three formation-mean ChRM directions for the Dianzhong (~64–60 Ma), Nianbo (~60–50 Ma), and Pana (50–44 Ma) formations yield positive fold tests with optimal data grouping around 100% of complete unfolding. Three corresponding paleopoles, at 66.4°N, 262.5°E (A_{95} = 6.3°), 69.7°N, 286.6°E (A_{95} = 6.3°), and 68.6°N, 232.3°E (A_{95} = 5.9°), all have ASDs of the VGPs compatible with the global PSV model during the last 5 Ma thus indicating adequate sampling of the time-averaged geomagnetic field.

(2) The new paleopoles define low paleolatitudes of ~7°N (~64–60 Ma), ~10°N (~60–50 Ma) and ~13°N (~50–44 Ma) for the Lhasa Terrane during formation of the Linzizong Group suggesting small but systematic northward motion of ~30 mm/yr of the southern margin of the Asian continent. Since the Lhasa Terrane appears to have occupied low paleolatitudes of ~15°N during the Cretaceous (Section 6.1), southward motion of several hundred kilometers may have occurred during latest Cretaceous times in response to marginal southerly motion of Eurasia.

(3) The updated paleopole computed from 18 sites in the Pana Formation of the Linzizong Group also yields a low paleolatitude of ~13°N. In contrast 59 of 60 acceptable site-mean observations from several studies in this formation suggest

a mean paleolatitude of ~23°N. Evaluation of this paleolatitudinal enigma indicates that it could result from large PSV, inclusion of transitional directions and uneven sampling but it requires further investigation of a wider sample for resolution.

(4) The India–Asia collision is dated at least prior to ~50 Ma given that no paleomagnetically-detectable crustal shortening or thickening was occurred within the stable Indian Craton. In the light of paleomagnetic constraints on the size of early Paleogene Greater India (Patzelt et al., 1996; Yi et al., 2011), it is concluded that initial contact between India and Asia occurred no later than ~60.5 ± 1.5 Ma.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.jseaes.2014.09.007.

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